Stanford HEALTH CARE
STANFORD MEDICINE

Stanford Health Care
Anatomic Pathology & Clinical Laboratories
[redacted]
Laboratory Director: [redacted]

PATIENT INFORMATION

MRN	Last Name	First Name	Sex	DOB
[redacted]	Bindal	Yashdeep	Male	[redacted]

SPECIMEN INFORMATION

Requisition Number	Diagnostic Summary	Date Collected	Date Received
[redacted]	Involved by adenocarcinoma	1/15/2025	1/20/2025

Specimen Site	Specimen Type	Pathology Case Number	Block ID
Liver (biopsy)	Formalin-fixed paraffin embedded tissue (FFPE)	[redacted]	[redacted]

Percent Tumor	Assay Performed	Pipeline Version
60%	STAMP - Stanford Actionable Mutation Panel for solid tumors (201 genes)	STAMP v4.8.0

ORDERING PHYSICIAN INFORMATION

Last Name	First Name	Middle Name
[redacted]		

SUMMARY OF FINDINGS

TERT Promoter mutation (Pathogenic)

Estimated tumor mutation burden: 7.0 mut/Mb†

The following variants are not known to be clinically relevant at this time:

APC P2086A (Unknown significance)
BRD4 M239L (Unknown significance)
CREBBP Q2216del (Unknown significance)
PDGFRA V367M (Unknown significance)
PTCH1 V1083M (Unknown significance)
BCR N1166S (Unknown significance)
CREBBP Q2214_Q2216dup (Unknown significance)
FGF3 V225I (Unknown significance)
PPP2R1A K280N (Unknown significance)

Please see the following pages for variant annotations.

†The estimated tumor mutation burden is for research use only.

Comment:

Atleast copy number gain in part of chromosome 1, and equivocal copy number losses in ARID1A, CDKN2C, NRAS, PIK3CA, TP63 and U2AF1 genes are detected, and reported for research use only (RUO). Confirmatory testing with an alternative method is recommended if clinically indicated.

Interpreted By [redacted]

Approved Date 1/31/2025

Stanford Health Care
Anatomic Pathology & Clinical Laboratories
[redacted]
Laboratory Director: [redacted]

VARIANTS

VARIANT	AA CHANGE	VAF	TRANSCRIPT	PATHOGENICITY
TERT c.-124C>T (p.?) chr5:g.1295228G>A		38%‡	NM_198253.3	Pathogenic

The c.-124C>T variant results in a cytosine to thymine change in the TERT gene promoter, 124 base pairs upstream of the TERT start site. TERT encodes telomerase reverse transcriptase, the catalytic subunit of the telomerase enzyme. Tumor-associated mutations in the TERT promoter occur most frequently at positions 124 and 146 upstream of the TERT start site and are referred to in the literature as C228T and C250T, respectively. These specific mutations lead to enhanced telomerase expression. TERT promoter mutations are found in many cancer types, with the highest frequencies reported in brain, urothelial, thyroid, hepatocellular, and skin cancers. References: PMID 23348506, 23887589, 24657534, 26941407.

TUMOR MUTATION BURDEN

Estimated tumor mutation burden: **7.0** mutations per megabase.

The reported tumor mutation burden (TMB) is an estimate of the overall number of nonsynonymous exonic mutations per megabase. Factors that affect estimates of TMB include type of tumor, removal of predicted germline variants, number of mutations, and identity of mutated genes. TMB is reported for research use only (RUO) on specimens with minimum 20% tumor.

VARIANTS OF UNKNOWN SIGNIFICANCE

VARIANT	AA CHANGE	VAF	TRANSCRIPT	PATHOGENICITY
PPP2R1A c.840G>C (p.Lys280Asn) chr19:g.52719064G>C	K280N	57%‡	NM_014225.6	Unknown significance
BCR c.3497A>G (p.Asn1166Ser) chr22:g.23656194A>G	N1166S	55%‡	NM_004327.4	Unknown significance
BRD4 c.715A>C (p.Met239Leu) chr19:g.15376299T>G	M239L	50%‡	NM_058243.3	Unknown significance
APC c.6256C>G (p.Pro2086Ala) chr5:g.112177547C>G	P2086A	48%‡	NM_000038.6	Unknown significance
FGF3 c.673G>A (p.Val225Ile) chr11:g.69625120C>T	V225I	48%‡	NM_005247.4	Unknown significance
PTCH1 c.3247G>A (p.Val1083Met) chr9:g.98218617C>T	V1083M	47%‡	NM_000264.5	Unknown significance
PDGFRA c.1099G>A (p.Val367Met) chr4:g.55133886G>A	V367M	44%‡	NM_006206.6	Unknown significance
CREBBP c.6633_6641dup (p.Gln2214_Gln2216dup) chr16:g.3778418_3778426dup	Q2214_Q2216dup	24%‡	NM_004380.3	Unknown significance
CREBBP c.6624_6626del (p.Gln2216del) chr16:g.3778427_3778429del	Q2216del	8%‡	NM_004380.3	Unknown significance

Stanford Health Care
Anatomic Pathology & Clinical Laboratories
[redacted]
Laboratory Director: [redacted]

Method:

‡STAMP reports include a variant allele fraction (VAF) for single nucleotide and insertion-deletion variants. This number represents the fraction of the total reads showing the variant. Observed VAF is affected by a number of factors, most importantly tumor purity (a low purity specimen will have low VAFs for somatic mutations), but also copy number (a mutated allele with high level amplification will have a high VAF), as well as assay-specific factors. Therefore, clear inferences regarding whether variants are germline or somatic, or whether some variants are subclonal cannot be made with high confidence.

This test covers 201 genes, either in part or fully, at a minimum analytic detection limit of 5%.

This assay detects potentially clinically actionable mutations, as well as additional genes that are frequently mutated in cancers. The Stanford Actionable Mutation Panel (STAMP) for solid tumors is a targeted next generation sequencing method that uses target enrichment to capture genomic regions of interest. The targeted sequencing approach and integrated bioinformatics workflow is optimized for ultra-deep sequencing of formalin fixed tumor biopsy tissue specimens. The workflow includes acoustic shearing of isolated genomic DNA, followed by efficient preparation of sequencing libraries and a target enrichment approach to capture genomic regions of interest for sequencing. The enrichment is done using custom designed libraries of capture oligonucleotides that target a specific set of genomic regions. This panel targets 201 genes, either in part or fully, with the genes selected on the basis of their known impact as actionable targets of existing and emerging anti-cancer therapies, their prognostic features, and/or their mutation recurrence frequency across patients with known cancer types. These genomic features are interrogated to achieve a minimum analytic detection-limit of at least 5%. Pooled libraries are sequenced on the Illumina NextSeq system. Due to inherent limitations of the NGS method, in general, insertion-deletion variants larger than 25 bp are not reliably detected.

This test is not intended to diagnose inherited diseases, and reported variants are only annotated for their relevance to cancer. If there is clinical suspicion for a non-cancer related inherited disease involving a gene on this panel, separate testing of a germline specimen is suggested.

For a full gene list, go to www.stanfordlab.com/content/stanfordlab/en/test-details/s/STAMPT.html

Genomic positions are given in reference to the GRCh37 (hg19) assembly of the human genome.

Disclaimer:

Molecular diagnostic test results should be interpreted in the context of standard clinical, laboratory and pathological findings. Molecular genetic test results impart a probabilistic risk of disease. In order to derive the most meaningful benefit from this testing, it is recommended that the results and subsequent options from these complex tests be discussed with patients by a trained genetics professional.

False negative results may be due to sampling error or errors in sample handling, as well as clonal/signal density below the level of detection.

Fusion calling for this assay has high specificity, but limited sensitivity. Clinical and pathologic correlation is recommended.

This test was developed and its performance characteristics determined by Stanford Clinical Laboratories. It has not been cleared or approved by the U.S. Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 ('CLIA') as qualified to perform high complexity clinical laboratory testing.

Resulting Lab

Hillview

Hillview Laboratory, [redacted]